



Technical Service Bulletin

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New Crankshaft

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin announces the release of a new fully counterweighted crankshaft for QSK60 industrial engines.

The fully counterweighted crankshaft has sixteen counterweights (Figure 1), one on each web, as opposed to the partially counterweighted crankshaft, which has eight counterweights (Figure 2).

The fully counterweighted crankshaft will aid in reducing the higher internal loads experienced within industrial applications, improving main bearing oil film thickness and main bearing life.

The fully counterweighted crankshaft will **only** be installed in industrial engines from production plants, but can be installed in most other applications in service. The **only** engines the fully counterweighted crankshaft can **not** be installed into are the Marine certified engines, because of Marine certification requirements.

The new fully counterweighted crankshaft went into partial production at Daventry Engine Plant, from 9 November 2009, and full, industrial, production on 11 January 2010. Engine serial number (ESN) first for full production in industrial applications is ESN 33180121. The following ESN's were built with the new fully counterweighted crankshaft between 9 November 2009 and 11 January 2010:

331795 55	331797 30	331797 97	331798 36	331798 59	331799 88	331800 01	331800 13	331800 36
331795 57	331797 32	331798 22	331798 45	331798 62	331799 89	331800 02	331800 25	331800 37
331795 95	331797 35	331798 27	331798 47	331798 65	331799 91	331800 03	331800 27	331800 49
331797 13	331797 45	331798 34	331798 56	331798 66	331799 92	331800 11	331800 28	
331797 15	331797 96	331798 35	331798 58	331799 87	331799 99	331800 12	331800 35	

Parts distribution centers will **only** stock the fully counterweighted and the Marine certified, partially counterweighted crankshafts.

QSK60 blocks have had a machine cut added to the left bank pan rail inner wall section at the main bearing locations (Figure 3). QSK60 blocks that do **not** have these machine cuts, will need to have some counterweights removed in order to remove and install the number 8 main bearing cap. Use the following procedures for further information for removing the main bearings and crankshaft. The procedures can be found in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-006-tr.html) Refer to Procedure 001-016 in Section 1. The machine cuts (Figure 3, Enlarged Area) can be added to older QSK60 blocks so the main bearing caps can be removed without removing the crankshaft counterweights (/qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html).

Note : The new QSK60 fully counterweighted crankshaft can be installed in all versions of QSK60 blocks.

Table 1: Fully Counterweighted Crankshaft (Industrial applications)		
Description	Old Part Number	New Part Number
Crankshaft (Assembly)	3089797	3642870

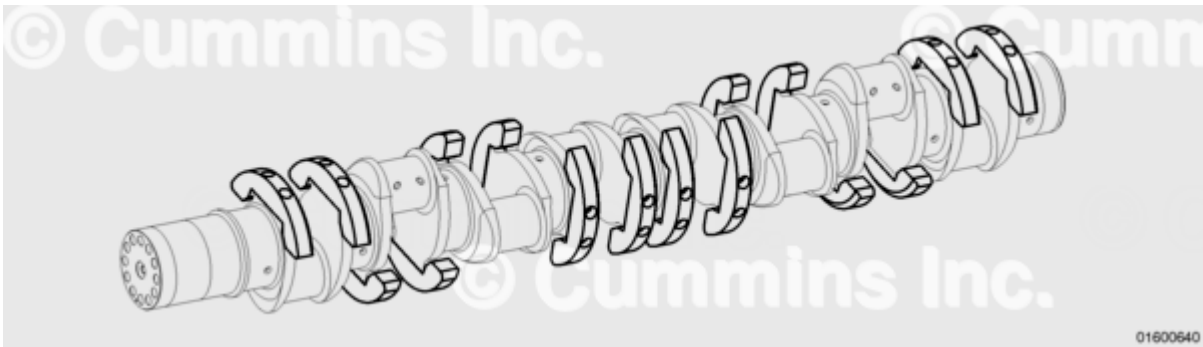


Figure 1 - QSK60 fully (16) counterweighted crankshaft

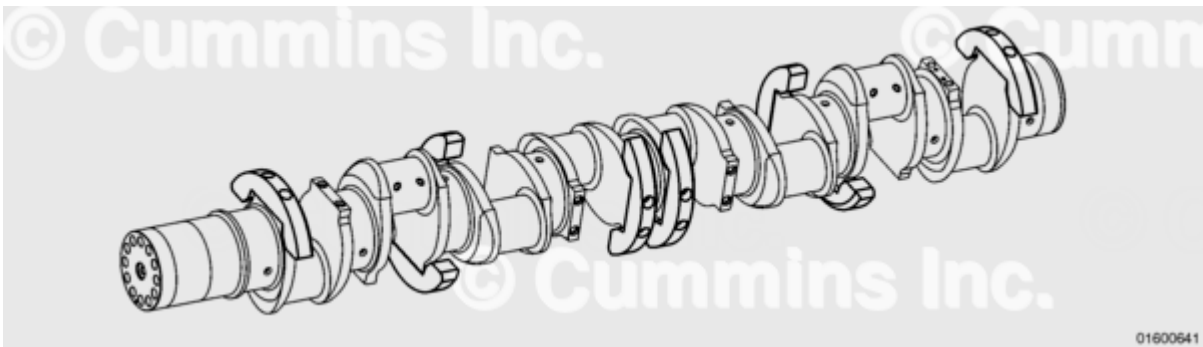


Figure 2 - QSK60 partially (8) counterweighted crankshaft

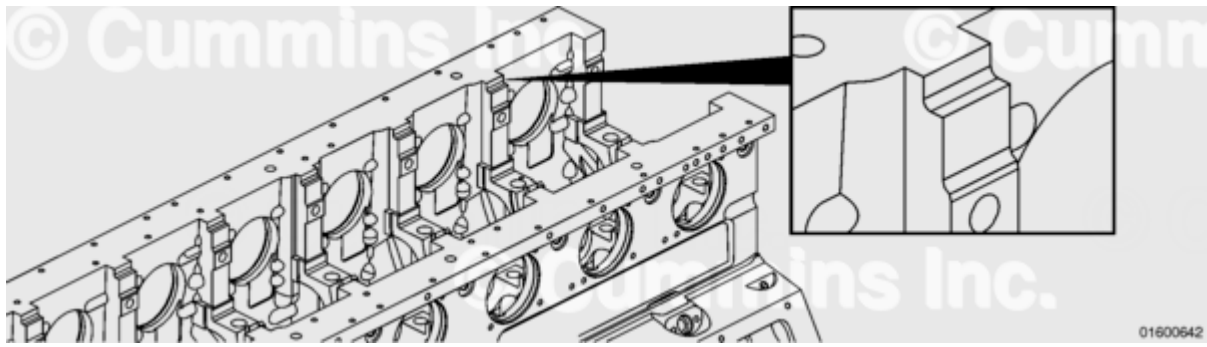


Figure 3 - Cylinder block showing additional machine cuts to the left bank pan rail inner wall section at the main bearing locations.

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